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Implementation and performance of the Neural-Ringer electron selection algorithm in the high-level trigger of the ATLAS experiment

Identifying electrons at the trigger level is crucial for many analyses in the ATLAS experiment. The instantaneous LHC luminosity increase from 2017 to 2018 resulted in up to 60 proton-proton interactions per bunch-crossing and challenged the electron trigger system. Trigger algorithms were then optimized to maintain high efficiency and a low background rate in this harsh environment. The Neural-Ringer algorithm was a part of this optimization in the high-level trigger, selecting electrons with a transverse energy above 15 GeV. Based on a set of multi-layer perceptron models, it is the first neural network employed as a baseline algorithm in the ATLAS trigger system. This algorithm resulted in an early rejection rate 13 times higher than the previous cut-based algorithm, with minimal loss in electron detection efficiency.

Analysis Team

[*email:* atlas-TRIG-2021-01-analysis-team@cern.ch]

Jose Seixas, Denis Oliveira Damazio, Chris Meyer, Werner Spolidoro Freund, Joao Victor Da Fonseca Pinto, Micael Verissimo De Araujo, Edmar Egidio Purcino de Souza, Eduardo Furtado de Simas Filho

Editorial Board

[*email:* atlas-TRIG-2021-01-edboard-conveners@cern.ch]

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Fernando Monticelli (La Plata) (chair)

Implementation and performance of the Neural-Ringer electron selection algorithm in the high-level trigger of the ATLAS experiment

The ATLAS Collaboration

Identifying electrons at the trigger level is crucial for many analyses in the ATLAS experiment. The instantaneous LHC luminosity increase from 2017 to 2018 resulted in up to 60 proton-proton interactions per bunch-crossing and challenged the electron trigger system. Trigger algorithms were then optimized to maintain high efficiency and a low background rate in this harsh environment. The Neural-Ringer algorithm was a part of this optimization in the high-level trigger, selecting electrons with a transverse energy above 15 GeV. Based on a set of multi-layer perceptron models, it is the first neural network employed as a baseline algorithm in the ATLAS trigger system. This algorithm resulted in an early rejection rate 13 times higher than the previous cut-based algorithm, with minimal loss in electron detection efficiency.

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1 Introduction

The ATLAS physics programme relies on an efficient trigger system [1], allowing the experiment to record highly signal-enriched subsets of collision events produced by the Large Hadron Collider (LHC) at CERN [2]. Electron triggers [3] are important for searches for new particles[4, 5], measurements of Standard Model cross-sections and properties of fundamental particles such as the Higgs [6, 7], Z and W bosons.

This paper describes the performance improvements achieved by the ATLAS trigger system thanks to the introduction of a neural network based algorithm, the Neural-Ringer, to identify electrons, while rejecting QCD backgrounds efficiently. The Neural-Ringer became the baseline algorithm for selecting isolated electrons online by the ATLAS trigger system in the 2017 proton–proton collision at $\sqrt{s} = 13$ TeV in LHC Run 2.

The ATLAS detector and the trigger system are described in section 2. Electron identification, both offline and online, is described in section 3. Full details of the Neural-Ringer electron identification, including the training strategies, are given in section 4. Online performance of the Neural-Ringer and its comparison with respect to the previous trigger baseline algorithm are described in section 5. A detailed statistical analysis of the Neural-Ringer electron results for electron identification is presented in section 6. Finally, the conclusions and prospects are presented in section 7.

2 Detector and Data acquisition

2.1 The ATLAS detector

ATLAS [8–10] is a multipurpose detector designed to characterize the particles produced in high-energy proton–proton (pp) and heavy-ion (HI) collisions at the LHC. It is composed of a tracking system in the innermost region around the interaction point, surrounded by calorimeters and muon chambers. The inner tracking detector (ID) is immersed in a 2 T magnetic field produced by a thin superconducting solenoid, and provides precise reconstruction of charged-particle tracks in a pseudorapidity¹ range $|\eta| < 2.47$. Its innermost part consists of a silicon pixel detector surrounded by a silicon microstrip tracker, typically with four sensor layers. The outermost region of the tracker is covered by a transition radiation tracker (TRT). It consists of straw drift tubes filled with a gas mixture of Xe, CO₂, and O₂ interleaved with polypropylene/polyethylene transition radiators, creating transition radiation for particles with a large Lorentz factor.

Surrounding the inner tracking system, the ATLAS detector features a calorimeter system composed of both electromagnetic (EM) and hadronic (HAD) sections. This system is designed to ensure complete ϕ coverage and spans the pseudorapidity range $|\eta| < 4.9$, with enhanced granularity in the region corresponding to the inner detector coverage.

¹ ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the center of the detector and the z -axis along the beam-pipe. The x -axis points from the IP to the centre of the LHC ring, and the y -axis points upward. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the beam-pipe. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$. The angular distance ΔR is defined as $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$. Transverse momenta and energies are defined as $p_T = p \sin \theta$ and $E_T = E \sin \theta$, respectively.

The EM calorimeter is a lead/liquid-argon (LAr) sampling calorimeter with an accordion-shaped geometry. It is segmented into two half-barrels ($-1.475 < \eta < 0$ and $0 < \eta < 1.475$) and two endcap modules ($1.375 < |\eta| < 3.2$). The transition area between the barrel and endcap sections ($1.37 < |\eta| < 1.52$) includes a considerable amount of inactive material [8]. Within the region where electrons are measured with precision ($|\eta| < 2.5$, excluding the transition region), the EM calorimeter is structured into three longitudinal layers along the shower depth. Each layer is composed of a set of calorimeter cells with a rectangular shape in the η, ϕ plane. The first layer features finely segmented cells in the η direction, which allow effective discrimination between isolated photons and closely spaced photon pairs resulting from $\pi^0 \rightarrow \gamma\gamma$ decays. For high transverse energy electrons and photons, the majority of energy is deposited in the second layer, with a cell granularity of 0.025×0.025 in (η, ϕ) space. The third layer captures energy deposits in the shower's tails. A thin presampler layer, located in front of the accordion calorimeter and covering $|\eta| < 1.8$, serves to correct for energy lost before reaching the calorimeter.

Beyond the EM calorimeter, three layers of hadronic calorimeters provide additional measurement capabilities. Additional background rejection for electrons and photons is achieved by detecting associated hadronic energy. The barrel hadronic calorimeter, covering $|\eta| < 1.7$, is a sampling calorimeter using iron and scintillator tiles, with light collection via wavelength-shifting fibers. The hadronic endcaps use copper/LAr calorimeters. In the forward region (FCal), copper–tungsten/LAr calorimeters are employed for both electromagnetic and hadronic measurements extending up to $|\eta| = 4.9$. All LAr-based calorimeters are enclosed within one barrel and two endcap cryostats.

The outermost component of ATLAS is the muon spectrometer (MS), which operates in the pseudorapidity range $|\eta| < 2.7$ and consists of three large toroidal magnet systems, each comprising eight coils. The magnetic field integral of the toroid system ranges from 2.0 to 6.0 T across most of its coverage. The MS integrates precision tracking chambers complemented by fast-response trigger detectors.

2.2 The trigger system

A hierarchical two-level trigger system [1] is used to select events of interest produced in LHC beam collisions. The first-level (L1) trigger, implemented with custom-made electronics, reads the calorimeters at a coarser granularity to reduce the event rate from the 40 MHz bunch-crossing rate to below 100 kHz. About 35% corresponds to events with high-energy electromagnetic showers, potentially originating from electrons or photons. The L1 trigger also defines regions-of-interest (RoIs), situated around local regions with high transverse energy, which are processed by the high-level trigger (HLT).

The HLT operates in a large computing farm and reduces the rate of events written to disk to an average of approximately 1 kHz. The HLT uses full-granularity calorimeter information, precision measurements from the muon spectrometer, and tracking information from the ID, which are not available at L1. HLT reconstruction can be executed either using cells within the RoIs identified at L1 or at full detector scale. The selection of particle candidates by the HLT is performed in a sequence of steps. First, a set of fast reconstruction algorithms is performed, using only detector information within the RoI, and faster and simpler algorithms to achieve early background rejection. Secondly, a set of offline-like precision reconstruction algorithms is executed at a lower rate, with access to the data from the entire detector.

A sequence of L1 and HLT trigger algorithms is called a “trigger” and is meant to identify one or more particles of a given type associated to a set of transverse energy or momentum thresholds to provide the trigger system final decision. Electron triggers are meant to select events with electrons in the final state identified by the detector. The “trigger menu” defines the configuration of the trigger system, setting a full

list of the L1 and HLT triggers. Menu composition and its corresponding trigger thresholds are optimised for the LHC running conditions (beam type, luminosity, etc.) to fit the event acceptance rate and the bandwidth constraints of the data acquisition system of the ATLAS detector, as well as the offline storage constraints.

3 Electron Identification

3.1 Offline object reconstruction and identification

The offline reconstruction of electrons and photons [11] is based on dynamic, variable-sized clusters of energy deposits, known as topo-clusters, formed from topologically connected electromagnetic (EM) and hadronic calorimeter cells [12]. These clusters help to recover energy lost due to bremsstrahlung photons or electrons originating from photon conversions. After applying initial corrections for position and energy calibration, the topo-clusters are matched to tracks in the ID that have been re-fitted to account for bremsstrahlung effects, following the method outlined in Ref. [13], to reconstruct electron candidates. Topo-clusters that are either unmatched to any track or matched to a track associated to a conversion vertex are reconstructed as photon candidates. Once candidates are formed, their energies undergo further recalibration for use in analyses.

Prompt electrons traversing the central region of the detector ($|\eta| < 2.47$) are identified using a likelihood-based (LH) method, which uses several discriminating variables: the longitudinal and lateral shapes of EM showers, the quality of the track, matching criteria between track and cluster, and particle identification information from the TRT. The probability density functions (pdfs) used in the LH approach are derived from $J/\psi \rightarrow ee$ decays for the transverse energy (E_T) range of 4.5–15 GeV, and from $Z \rightarrow ee$ decays for $E_T > 15$ GeV, as detailed in [11]. Separate pdfs are defined for each discriminant variable across different bins of E_T and η of the electron-candidate.

Three identification operating points (called “loose”, “medium” and “tight”) are defined, each corresponding to progressively tighter requirements on the LH discriminant. The average efficiencies for electrons from electroweak processes (evaluated over the 2015–2017 dataset) are approximately: 93% (loose), 88% (medium), and 80% (tight).

3.2 Online reconstruction and identification at trigger level

Electron reconstruction at the HLT stage is performed on each EM RoI provided by L1, which satisfies E_T and isolation requirements as specified by the trigger configuration. It proceeds in a series of sequential steps as shown in the schematic diagram in Figure 1. Lozenges represent feature-extracting algorithms that are responsible for building electron properties from detector data (calorimeter shower shapes, transverse momentum, energy, position, etc.). Rectangles represent hypothesis-testing algorithms, which apply selection requirements on those reconstructed properties.

Electron identification at the HLT selection begins with a fast calorimeter reconstruction (FastCalo) step, which has two implementations: the original/baseline cut-based selection and the novel neural-network (NN) approach using the Neural-Ringer algorithm presented in this paper. The latter is implemented for triggering electrons with $E_T \geq 15$ GeV. FastCalo is followed by a fast tracking-reconstruction step, which builds tracks for electron candidates only inside the RoI. A fast electron hypothesis algorithm applies

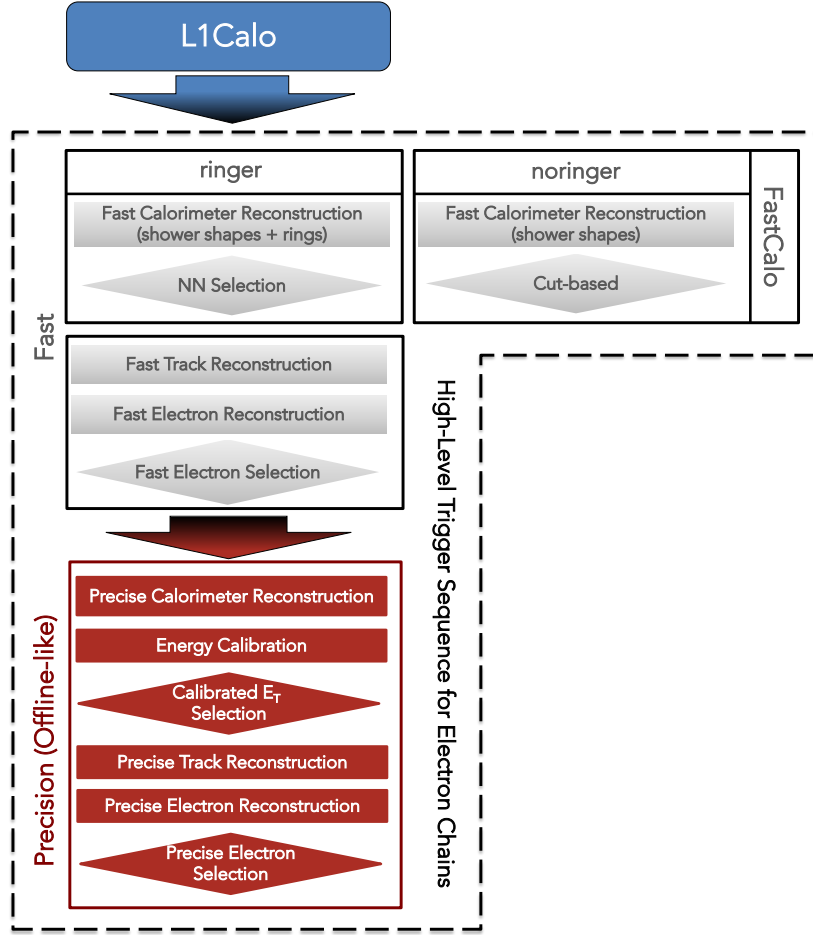


Figure 1: Diagram of the sequence of algorithms used to select electron candidates at the High Level Trigger stage. Differences between the original baseline (no-ringer) and ringer-based electron triggers lie in the first step, called FastCalo. Grey blocks represent algorithms in the fast stage, and red blocks represent precision-step algorithms.

identification requirements by combining the results of both fast algorithms. Reconstruction algorithms are then executed for trigger electron candidates satisfying the fast electron hypothesis selection.

In the precision calorimeter reconstruction step, the cluster reconstruction and calibration are similar to those for photons, and to those used offline in early Run 2 analyses [14]. Then, ID tracks within the RoI are extrapolated to the second layer of the EM calorimeter and are required to match the electron candidate cluster within $|\Delta\eta(\text{track}, \text{cluster})| < 0.05$ and $|\Delta\phi(\text{track}, \text{cluster})| < 0.05$ rad. The precision electron selection combines the results of the precision calorimeter and precision tracking steps to build a likelihood (LH) discriminant with four operating points: “lhvloose”, “lhloose”, “lhmedium”, and “lhtight”. The identification in the trigger is designed to be as close as possible to the offline version, except for the following differences: the discriminating variables used online have different resolutions; the momentum loss due to bremsstrahlung, $\Delta p/p$, is not accounted for in the online LH.

An important aspect for trigger operation is the amount of simultaneous particle interactions in the detector (μ), which generates pileup of events in the electronics readout system and may change the typical energy deposition profile in the calorimeters for electromagnetic particles. The online selection uses an estimate

of $\langle\mu\rangle$, the mean number of collisions during a reference period, to measure pileup instead of the number of primary vertices used offline.

An additional, optional requirement of isolation denoted ‘ivarloose’ is also available for electron triggers. This tracking-only isolation is required to satisfy $p_T^{iso}(\Delta R^{var} < 0.2)/p_T < 0.10$ and is calculated similarly to the offline isolation operating points.

4 The Neural-Ringer algorithm

The identification of electromagnetic objects at the FastCalo stage of the HLT must efficiently reject the large hadronic background while operating under tight latency constraints. The Neural-Ringer algorithm was developed to enhance this discrimination capability by exploiting the symmetry of electromagnetic showers in the calorimeter. Electrons and photons deposit most of their energy in a compact, centrally concentrated region, whereas hadronic jets exhibit broader, more irregular energy distributions extending into the hadronic calorimeter.

The idea behind the Neural-Ringer algorithm is to draw concentric rings perpendicular to the particle axis and estimate the energy deposited along each ring. This way, the algorithm summarizes the shower profile as a function of the radial distance from the shower axis, converting the spatially distributed calorimeter information into a compact, physics-motivated representation suitable for neural network classification.

4.1 Building the rings

The implementation of the Neural-Ringer defines rings at each layer of both EM and HAD calorimeters. In the ATLAS calorimeter, the rings are built with a rectangular shape to match the grid layout of the calorimeter cells. Since this algorithm was designed to be used in the FastCalo step of the Trigger reconstruction, all the rings are confined within the area defined by the RoI and centered around the cell with the highest energy deposited closest to the L1 RoI position. As a general rule, a ring of size n in a given layer of the calorimeter consists of the cells lying in the border of an $n \times n$ cell area.

The actual definition of the rings is depicted in Figure 2. In the EM calorimeter, rings are centered around the most energetic cell at each layer, while in the hadronic calorimeter, the $\eta \times \phi$ position of the most energetic cell in the second layer of the EM calorimeter is used to determine the central ring. Formally, the cells that belong to a given ring of size n on a given layer l is given by Equation 1 where $\eta_{i,l}$, $\phi_{i,l}$ represent the position of the cell i at detector layer l , the subindex “hot” refers to the central cell (first ring) from a given layer; $h_{\eta,l}$ and $h_{\phi,l}$ are the sizes in η and ϕ of the cell at the layer l ; $\Theta_{\text{RoI},l}$ is the set of cells in the l^{th} layer which are within the RoI; and $\lfloor \cdot \rfloor$ is the *round* function. Table 1 shows the number of rings computed at each layer and the respective granularity.

$$R_{n,l} = \left\{ c_{n,l} \mid n = \left\lceil \max \left(\frac{|\eta_{i,l} - \eta_{\text{hot},l}|}{h_{\eta,l}}, \frac{|\phi_{i,l} - \phi_{\text{hot},l}|}{h_{\phi,l}} \right) \right\rceil, \forall c_{i,l} \in \Theta_{\text{RoI},l} \right\}, \quad (1)$$

For every layer of the EM and HAD calorimeters, the transverse energy deposited in the cells of each ring is summed. Across all layers, a total of 100 ring transverse energies are computed, ordered from the innermost EM region to the outermost hadronic layer, and stored in a single vector. Each index in that vector identifies a ring. The energy normalization of the rings is performed as in Eq. (2) where r_k or r_i is

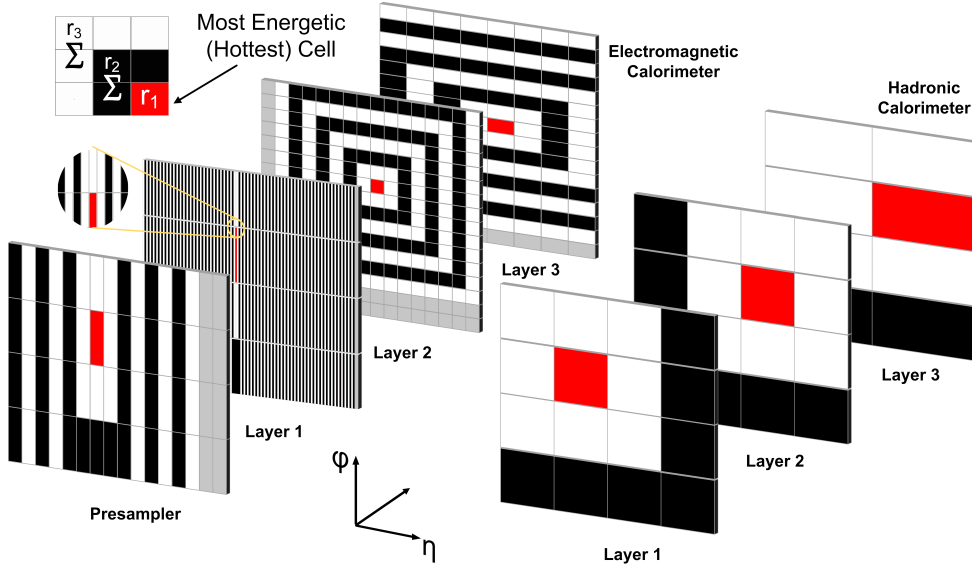


Figure 2: Sketch illustrating the ring-shaped energy description. The most energetic cell of the RoI is in red, while the consecutive neighbouring rings are represented by alternating white and black cells. Extracted from [3].

Table 1: Neural-Ringer parameters for the different calorimeter sampling layers and sections. The Neural-Ringer algorithm parameters are dependent on the layer l but independent of E_T . The reconstruction parameters are the cell size in η ($h_{\eta,l}$), ϕ ($h_{\phi,l}$) and the number of rings to be computed in each layer (N_l).

Ringer		Calorimeter Sampling			Parameters		
Section	Layer (l)	Barrel	ITC	End-cap	$h_{\eta,l}$	$h_{\phi,l}$	N_l
EM	PS	PreSamplerB		PreSamplerE	0.025	0.1	8
	EM1	EMB1		EMEC1	0.003	0.1	64
	EM2	EMB2		EMEC2	0.025	0.025	8
	EM3	EMB3		EMEC3	0.050	0.025	8
	HAD1	TileBar0 TileExt0	TileGap3	HEC0	0.1	0.1	4
HAD	HAD2	TileBar1 TileExt1	TileGap1	HEC1 HEC2	0.1	0.1	4
	HAD3	TileBar2 TileExt2	TileGap2	HEC3	0.2	0.1	4

the sum of transverse energy of the cells in the ring with index k or i , and r'_k is the normalized ring energy value with index k . The absolute value is used in the denominator to handle extreme cases where the total energy sum is negative. In this way, the consistency of positive or negative values is maintained after normalization. This procedure was initially proposed and examined in Ref. [15] as a way of preserving the shower energy profile as fractions of the total energy. Building rings in this way summarizes the information of thousands of calorimeter cells into just 100 variables, while preserving both the longitudinal and lateral structure of the energy deposition.

$$r'_k = \frac{r_k}{\sum_{i=1}^{100} r_i}, \quad \forall k \in \{1, \dots, 100\}, \quad (2)$$

Figure 3 shows the distribution of ring energy as a function of the ring number (or vector index) for electrons and QCD jets obtained in simulated events. Rings within vertical dashed lines belong to the same layer of the same calorimeter. The distribution of the ring energy for layer 2 of the EM calorimeter (EM2) is shown in the figure, zoomed in.

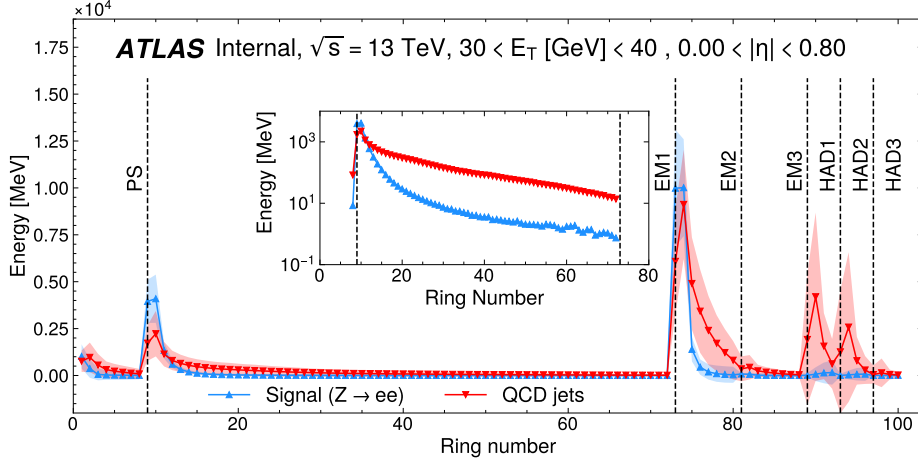


Figure 3: Simulated average ring transverse energy profiles for jets and electrons in the bin $30 \leq E_T < 40$ GeV and $0.0 \leq \eta < 0.8$ for each layer of the calorimeter. The profile in the inset shows the EM1-layer ring energies on a logarithmic scale. Almost all the energy of the electrons is absorbed by the electromagnetic layers, and no energy deposit is expected in the hadronic calorimeter. On the contrary, a significant fraction of the jet energy is absorbed by the hadronic layers, and only a part of it is absorbed by the electromagnetic calorimeter. The shaded area indicates the standard deviation of each ring energy.

4.2 Motivation for using a set of neural networks

The detector response and discrimination power between electrons and QCD jets vary with the candidate E_T and its η position, due to the different granularities of the calorimeters at different pseudorapidities. Therefore, a set of multi-layer perceptrons (MLPs) trained on different regions of $E_T \times |\eta|$ is used to classify the normalized vector of ring energies. Splitting the training into multiple small models enables higher discrimination power and efficient in terms of computing resources compared to a single model that is inclusive over all E_T and $|\eta|$ regions.

Low-complexity models, featuring a single hidden layer and a small number of neurons, were selected for the MPL set to accommodate limited CPU resources during the Fast Processing step of the Trigger and to minimize latency.

4.3 Datasets and training

The Neural-Ringer algorithm implemented online during 2017 was trained with simulated samples of $Z \rightarrow e^+e^-$ events for signal and QCD dijet events for background. To enrich the sample of electron backgrounds in the QCD simulation, only events where the sum of transverse energy of particles of the hard scatter (excluding muons and neutrinos) in a window of $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$ exceeds 17 GeV are

selected. Both $Z \rightarrow e^+e^-$ and hadronic jets from QCD hard-scatter simulated events included pileup of 60 interactions per bunch crossing, covering the full Run-2 running conditions.

The Neural-Ringer implementation deployed for 2018 was trained using data from LHC pp collisions at $\sqrt{S} = 13$ TeV collected during the 2017 run, that were approved in ATLAS collaboration requirements for beam stability, detector operation, and data quality [16]. A dedicated MLP model was defined for each $E_T \times |\eta|$ region.

Threshold values for the NN outputs were defined based on 2016 collision data, aiming for a signal efficiency comparable to that of the cut-based trigger. The training procedure and decision-making process were identical across all phase-space regions. Each MLP serves as an expert model corresponding to a specific phase-space region, with its own topology and parameters.

To train the models, electrons from the signal sample are selected using the tag-and-probe method described in [3], in which one electron (the tag) is used to trigger the event and is requested to satisfy tight identification and isolation requirements offline. A second electron (the probe) is required to be in the acceptance of the EM Calo, requiring only that the invariant mass of the Tag and Probe system is compatible with the Z boson mass window.

The model parameters were optimized using events selected from simulated $Z \rightarrow e^+e^-$ datasets. The network architecture consists of a single fully connected hidden layer with 5 to 20 hidden neurons, an input layer with 100 nodes (one for each ring), and an output layer with a single neuron. The activation function for both the hidden and output neurons is the hyperbolic tangent, which is commonly used in MLP designs [17].

To assess the statistical fluctuations in the efficiency measurements with respect to the model parameters obtained from the training procedure, the stratified k -fold cross-validation method [17] with $k = 10$ was employed. The stratified k -fold is one of the most common cross-validation techniques and consists of partitioning the dataset into k disjoint subsets, with each model being tested on the i^{th} subset after being trained on the other $k - 1$ subsets. The dataset stratification follows the empirical order of event selection, allowing for straightforward job parallelization; therefore, no random permutation is employed.

For each cross-validation sort and working point, 100 networks, initialized as in [18], are optimized by back-propagating the mean squared error through the RPROP algorithm [19]. This repeated model initialization aims to avoid suboptimal solutions that may arise from gradient descent algorithms. For each training procedure, only three models out of the 100 are retained: two for retrieving the best efficiency when fixing the sensitivity or false positive probabilities to the reference FastCalo efficiency, and another that gives the maximum SP value.

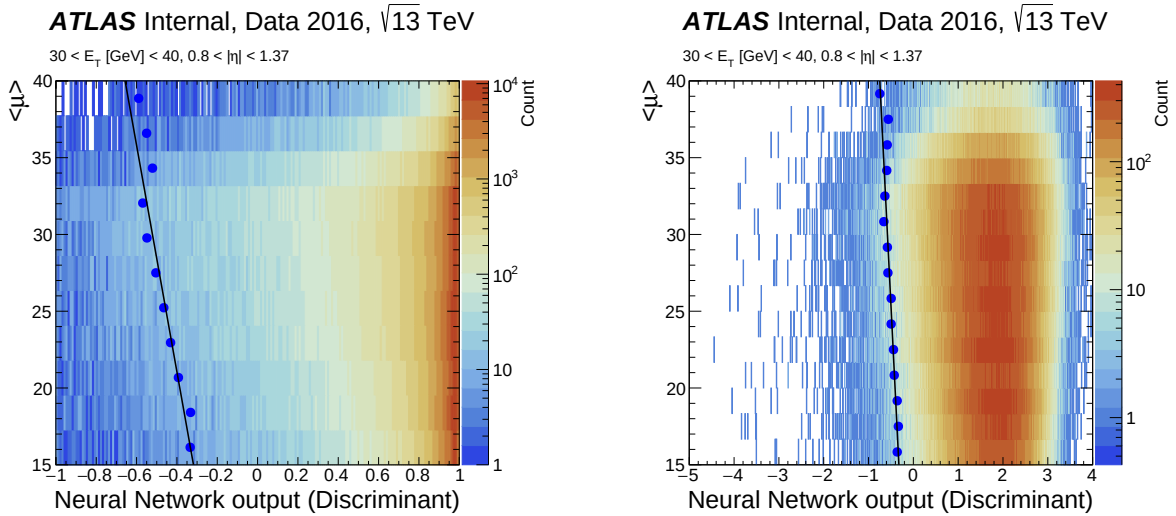
The maximum SP is a value that optimizes the balance between signal detection probability and the probability of fake acceptance in the ROC curve (i.e., the knee of the curve) and is defined as:

$$SP = \sqrt{\sqrt{P_D(1 - F_A)} \cdot \frac{P_D + (1 - F_A)}{2}}$$

where P_D is the signal detection probability (or sensitivity) and F_A is the probability of a fake signal (false positive). An early stopping technique [17] is employed to prevent overtraining. The selection of the optimal iteration and operating model is based on a heuristic approach (multi-stop), as described in [20].

The training strategy considered five regions in $|\eta|$ ($0.0 \rightarrow 0.8 \rightarrow 1.37 \rightarrow 1.57 \rightarrow 2.37 \rightarrow 2.47$) and five regions in E_T ($15 \text{ GeV} \rightarrow 20 \text{ GeV} \rightarrow 30 \text{ GeV} \rightarrow 40 \text{ GeV} \rightarrow 50 \text{ GeV} \rightarrow \infty$). To comply with stratified k-fold initializations and the search for the optimal number of hidden neurons (5-20), it was necessary to tune approximately 1.3 million neural networks. Finally, the process converged to a set of 25 MLPs for each of the four working points employed by electron triggers in the HLT.

After the training stage, particle identification is done by applying a threshold requirement to the one-dimensional output node of the MLP associated with the eta region of a given electron candidate. As in the HLT likelihood [21] algorithm, the threshold value used in the Neural-Ringer algorithm should be computed as a linear function of $\langle\mu\rangle$. This allows threshold adjustment as $\langle\mu\rangle$ changes, maintaining the detection rate approximately constant. However, a non-linear behaviour was observed near the asymptote of the output activation function (see Figure 4(a)) when using the ‘medium’ or ‘tight’ operating point. This was caused by the nonlinear activation function in the output neuron, and it can be avoided by replacing the hyperbolic tangent activation function, only in the output neuron, with a linear function (Figure 4(b)). All neurons in the hidden layer retained the hyperbolic tangent activation function, preserving the architecture used during training. This simple strategy ensures a linear relation of the targeted² signal efficiency with respect to the online pileup estimator, similar to what was used in [21]. To account for Monte Carlo and collision data differences (see Figure 5), the parameters of the linear threshold correction were derived using 2016 collision data and an upper bound of $\langle\mu\rangle = 40$ pileup interactions (i.e. the corrected threshold value is constant for pileup values above this upper bound).



(a) MLP output with a hyperbolic tangent activation function in the output neuron versus pileup.

(b) MLP output with a linear activation function in the output neuron versus pileup.

Figure 4: Neural-Ringer output as a function of pileup, $\langle\mu\rangle$, for $Z \rightarrow e^+e^-$ probes in 2016 collision data selected using the training criteria. The computed threshold per $\langle\mu\rangle$ unit for achieving the targeted efficiency is shown as a full blue circle. The black line is a linear fit to the thresholds. The output space is computed using the trained model without modifications in (a), while in (b), the activation function in the output neuron is replaced with a linear function.

² same signal detection probability as the online precision electron LH algorithm, increased by a factor of 50% for each of the four trigger operation points.

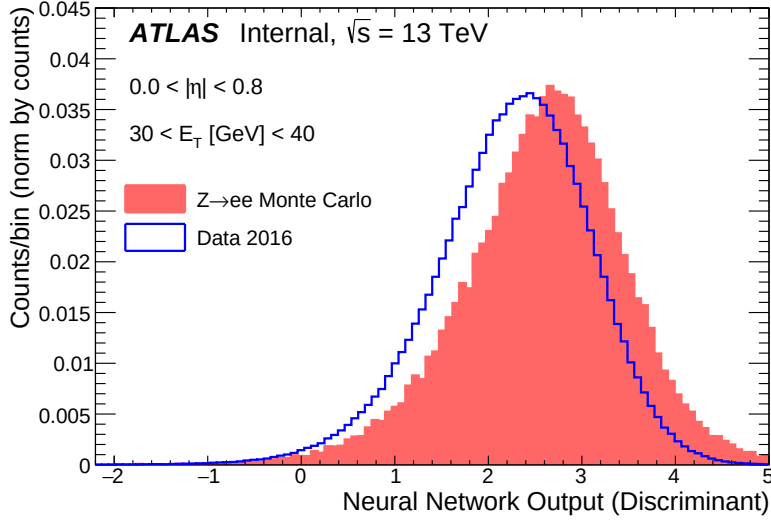


Figure 5: Differences between simulation samples (filled histogram) and collision data (empty histogram) obtained in 2016 $Z \rightarrow e^+e^-$ tag-and-probe for one region of the phase space. The shift in discriminant value between simulation and collision data is caused by differences between the simulation and data rings used as input to the model.

5 Online deployment and performance

The Neural-Ringer algorithm was implemented in three stages during 2017–2018 in LHC Run 2:

- i The commissioning stage in early 2017 runs (5.4 fb^{-1}), where all primary triggers were duplicated and both the Neural-Ringer and cut-based algorithm run in parallel in the FastCalo stage of the HLT.
- ii The operation stage, starting after Technical Stop 1 (TS1) in 2017. Triggers using Neural-Ringer were used as the main triggers, and a small set of cut-based triggers was kept active solely for validation. Final validation of the Neural-Ringer was performed by comparing the main triggers and the cut-based triggers from this period (39.0 fb^{-1}) as shown in Section 5.2 and Section 6.
- iii The Neural-Ringer-only stage, during 2018. Duplicated cut-based triggers were removed. The performance of the Neural-Ringer in 2018 is presented in Section 5.2. The results presented for this case are consistent with the second part of Run 2 and the full Run 3 scenario.

This section reports Neural-Ringer’s operational performance in each of these periods. In addition, the observed impact on CPU usage is discussed in Section 5.3.

5.1 Measurement of the electron trigger efficiency

The electron trigger efficiency is measured using the tag-and-probe method, which provides a sample of electrons selected without bias from the trigger decision. This technique allows unbiased estimation of the trigger performance by identifying probe electrons in events where a tag electron satisfies stringent offline and trigger requirements.

The efficiency is defined as the fraction of probe electrons that are accepted by the trigger, among those that pass the full offline selection:

$$\epsilon_{\text{trigger}} = \frac{N_{\text{trigger} \cap \text{offline}}}{N_{\text{offline}}} \quad (3)$$

where $N_{\text{trigger} \cap \text{offline}}$ is the number of probe electrons that pass both the offline selection and are accepted by the trigger, and N_{offline} is the total number of probe electrons passing the offline selection.

5.2 Assessment and deployment of Neural Ringer trigger Algorithm during Run 2 operation

The operation of Neural-Ringer commenced after the technical stop period (TS1) occurred in mid-2017. For monitoring purposes, a single-electron trigger was used. This trigger requires $E_T > 28$ GeV, with the selection criteria ‘tight’ and with ‘loose’ isolation criteria.

Signal efficiencies for both cut-based and Neural-Ringer triggers, evaluated for tag-and-probe electron candidates from 2017 data, are compared in Figure 6, where one can see both trigger efficiencies with respect to the offline Tight identification working point as a function of E_T , η and $\langle\mu\rangle$. The results show no significant differences, on the order of a few per mille, between the two algorithms. Although a slight efficiency loss is observed at higher $\langle\mu\rangle$ values in Figure 6(c) for the Neural-Ringer trigger, the overall electron efficiency was maintained nearly constant, consistent with the target efficiency of the Neural-Ringer algorithm’s training. The observed efficiency loss occurs after the application of the linear threshold correction at $\langle\mu\rangle = 40$ in 2017.

Other important results were assessed by comparing the trigger efficiencies in the 2017 collision data collected before and after switching to the Neural-Ringer algorithms. As shown in Figure 7, the Neural-Ringer signal detection efficiency was approximately the same as the cut-based selection for the three trigger setups evaluated. One of these triggers requires an electron candidate with $E_T > 17$ GeV, seeded by the first-level trigger with $E_T > 15$ GeV, and satisfying the loosest identification criteria (lhvloose). On the other hand, another requires an electron candidate with $E_T > 26$ GeV satisfying ‘tight’ identification (lhtight) with ‘loose’ isolation (ivarloose), and the last requires an electron candidate with $E_T > 60$ GeV satisfying ‘medium’ identification (lhmedium). The electron efficiency was kept nearly unchanged for these relevant single-electron triggers.

The presence of duplicated triggers, one configured with the Neural-Ringer algorithm and the other cut-based, allowed a direct comparison of their rates for fake electron candidates in collision data. With the Neural-Ringer, the overall fake-electron rate is reduced by a factor of 13.75 at the FastCalo step. It can be seen on the left side of Figure 8 that the improvement at the FastCalo step is similar for all regions in the evaluation with respect to E_T , η , and $\langle\mu\rangle$ variables, which is particularly interesting when considering the low E_T and endcap regions. Besides improving early fake-electron rejection, the use of the Neural-Ringer also resulted in a factor of two reduction in the final fake-electron rate from the HLT (see right side of Figure 8), mostly coming from the calorimeter’s barrel-endcap transition regions.

For the 2018 data-taking period, the Neural-Ringer was retrained using collision data, thereby improving electron identification performance. The signal efficiency of the lowest-transverse-energy-threshold unprescaled trigger increased by approximately 1%, primarily due to several minor enhancements in the full electron selection applied both online and offline. In parallel, a modest reduction in fake-electron

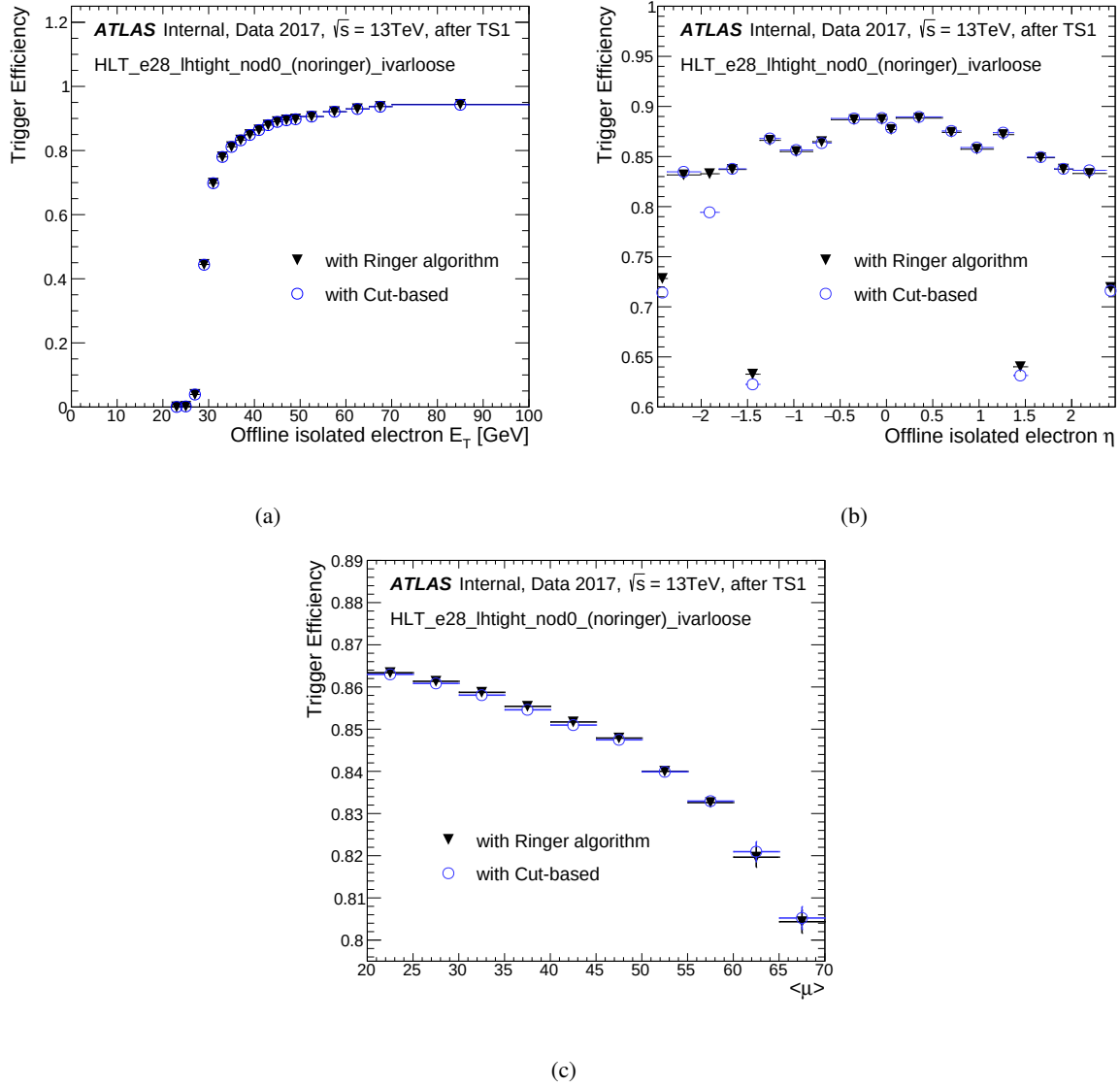


Figure 6: The efficiency of $E_T > 28$ GeV single-isolated-electron triggers with the Neural-Ringer algorithm and with the cut-based algorithms as a function of the offline electron (a) E_T , (b) η , and (c) $\langle\mu\rangle$ after its deployment in July 2017. Efficiency is given with respect to the offline Tight identification working point. For (b) and (c), only offline candidates with $E_T > 29$ GeV are used.

acceptance was observed at the FastCalo stage for certain unprescaled triggers relative to the 2017 post-TS1 configuration. Specifically, the lowest- E_T -threshold triggers ($E_T > 17, 26, 28$) achieved a reduction factor of 1.19, while the highest- E_T -threshold trigger ($E_T > 60$) reached a more significant reduction factor of 1.69.

5.3 Impact on CPU demands

The overhead required to compute the NN decision was evaluated by running the single-isolated-electron trigger requiring $E_T > 17$ GeV trigger with the Neural-Ringer algorithm or cut-based approach, in two

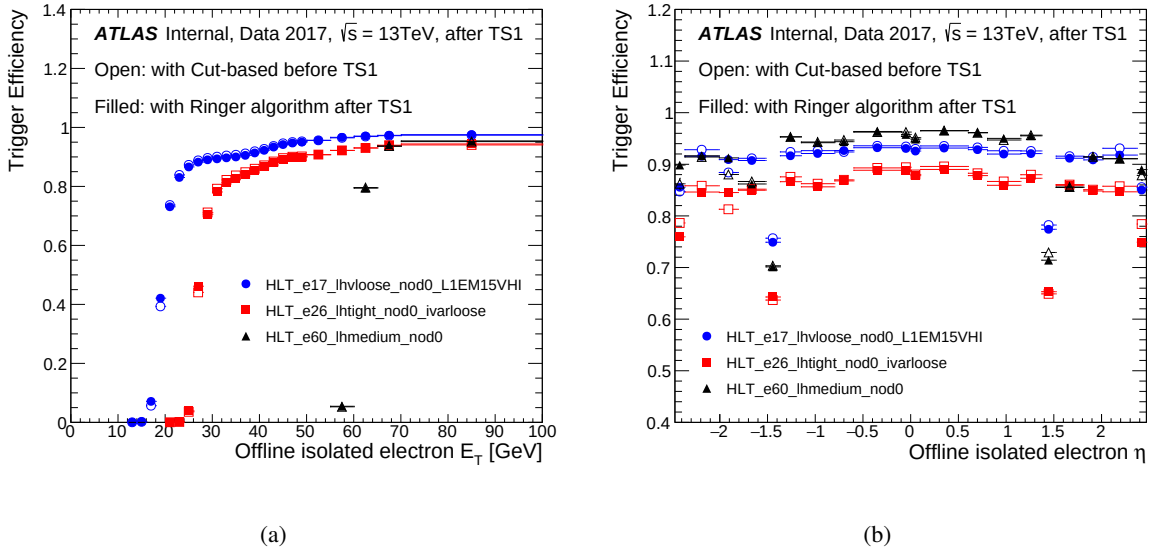


Figure 7: Efficiency of three single-electron triggers as a function of (a) E_T and (b) η for 2017 collision data. Open (filled) markers show the efficiency measurements in runs before (after) the deployment of the Neural-Ringer, thus referring to triggers executed without (with) the Neural-Ringer algorithm.

individual non-concurrent executions using the same dedicated computer system³ on the 2017 Enhanced bias [22] (EB) data. Hence, the measurements are performed under representative pile-up and topological conditions, with the reconstruction algorithm applied to multiple RoIs in the same bunch-crossing event.

To avoid a complete re-implementation of the FastCalo trigger step, electron triggers using Neural-Ringer information make their decision after cut-based features are reconstructed. In this case, the cut-based decision is not computed. This strategy also allows us to continue using cut-based chains for monitoring purposes. As a result, the Neural-Ringer electron triggers always require additional FastCalo processing time. Figures 9 and 10 show comparisons of the total processing time per call for the feature extraction and hypothesis-testing algorithms in the FastCalo step, respectively (see the related FastCalo blocks in Figure 1).

A multi-modal structure can be observed in the feature-extraction distributions (see Figure 9) for both trigger configurations, and it is associated with the algorithm for retrieving EM2 cell energies and building related shower-shape variables. The computation of the ring variables, which is the only difference between the triggers in the FastCalo feature extraction, requires an additional average processing time of 0.18 ± 0.03 ms/event.

A smaller contribution comes from hypothesis testing (see Figure 10), in which the ring variables are normalized, the discriminating variable's value is computed, and compared with the selection requirement. Hypothesis testing for the Neural-Ringer exhibits significantly higher computational demands than the cut-based selection, with processing times reaching up to 0.14 ms/event. This overhead is predominantly due to storing the neural network output, which requires 0.07 ± 0.02 ms per event. Despite this, the hypothesis testing remains less time-intensive than the feature extraction stage. Additionally, the time

³ This performance study was performed using a Xeon Phi 7120 (1.7 GHz, 32 threads) node with a memory of 256 GB@1333 running a SL6 Operating System.

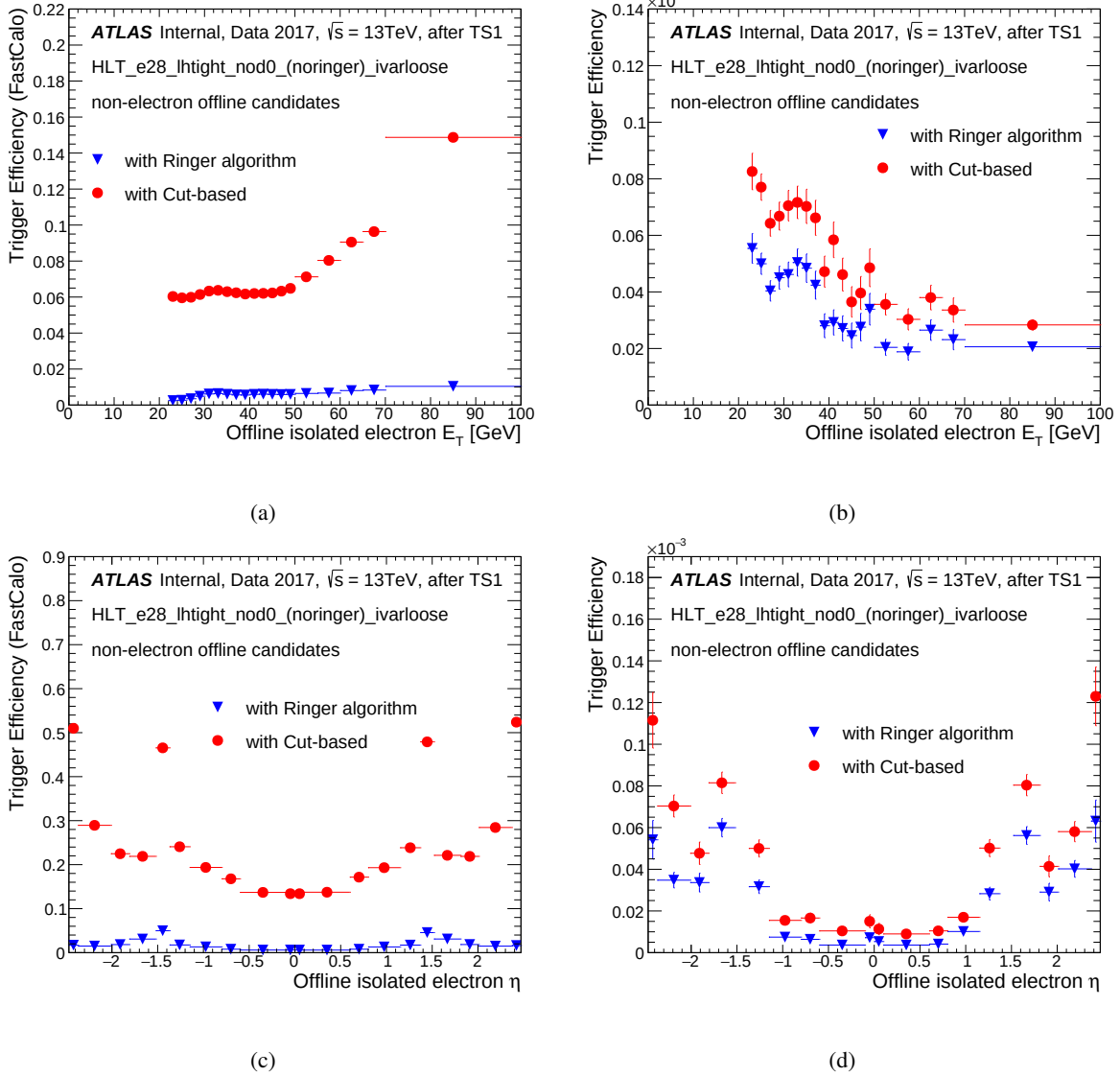


Figure 8: Efficiency of selecting background at the FastCalo (left) and at the final HLT decision (right) steps as a function of E_T (top) and η (bottom) with a single-isolated-electron trigger requiring $E_T > 28$ GeV and ‘tight’ selection, with and without the Neural-Ringer algorithm, employing 2017 collision data collected after the Neural-Ringer algorithm deployment.

increase is negligible in comparison with the average processing time of a single-electron trigger across the entire HLT (≈ 500 ms in Run 2).

Taking these values into account, the Neural-Ringer can require up to 50% more FastCalo CPU time per event than the trigger without the Neural-Ringer.⁴ Nonetheless, the Neural-Ringer provides a more discriminating selection, which reduces the number of CPU-demanding triggers by decreasing the need to process fake electrons in subsequent steps that are more computationally intensive. As shown in previous

⁴ The CPU-time increase is expected to depend on the trigger configuration because of the presence of pileup. The reported results are for the single-electron trigger, requiring $E_T > 17$ GeV, with a very loose operation point.

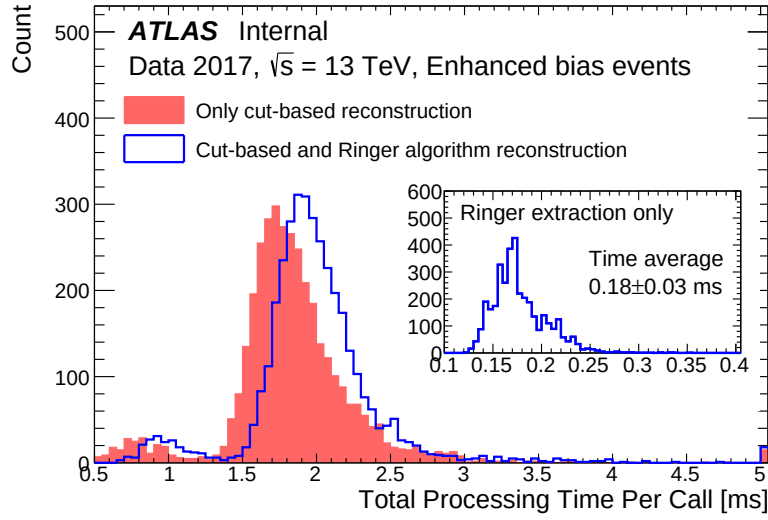


Figure 9: Total processing time per call for the feature extraction algorithms in the FastCalo step of an electron trigger requiring $E_T > 17$ GeV and Loose identification criteria, with the Neural-Ringer (blue) or the cut-based approach (red), on EB events (peak $\langle \mu \rangle = 45$). The inset on the right shows the processing time per call for the ring variable extraction only.

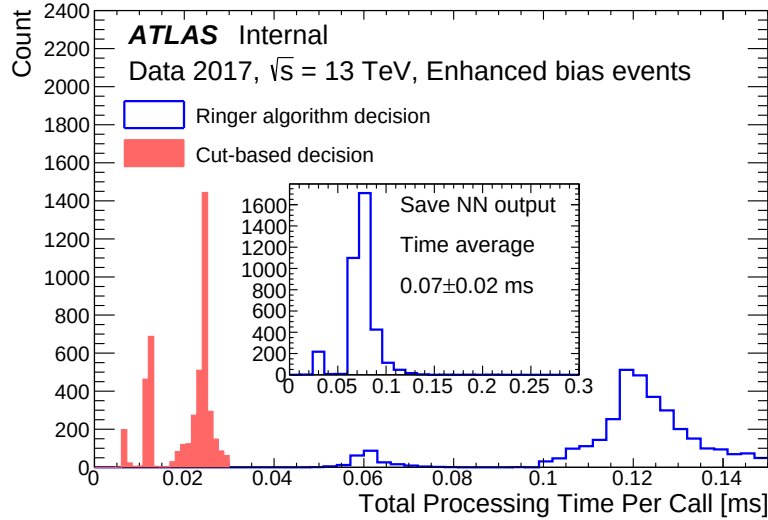


Figure 10: Total processing time per call for the hypothesis-testing algorithms in the FastCalo step of an electron trigger requiring $E_T > 17$ GeV and Loose identification criteria, with the Neural-Ringer (blue) or the cut-based approach (red), on EB events (peak $\langle \mu \rangle = 45$). The center histogram shows the processing time per call to store the entire Neural-Ringer output in a persistent file format.

373 results (see, for example, Fig 8), the overall fake-electron rate at the FastCalo step was reduced by a factor
 374 of 13.75, preventing these fake events from being processed in the further trigger steps.

6 Neural-Ringer vs cut-based triggers

This section presents a detailed comparison between the Neural-Ringer and the cut-based electron trigger algorithm used in the ATLAS experiment during the 2017 and 2018 data-taking periods. The study aims to identify differences in trigger decisions and to evaluate potential biases introduced by the Neural-Ringer within the electron trigger sequence.

The comparison is performed using a dataset from 2017, selected via the $Z \rightarrow e^+e^-$ tag-and-probe method [3]. The initial analysis applies the Neural-Ringer selection to the probe electron to assess trigger performance (Quadrant Analysis).

To further investigate algorithm-induced biases, a second dataset was collected during a dedicated 2017 run in which both the cut-based and Neural-Ringer algorithms were simultaneously active. The corresponding tag-and-probe sample enables a direct evaluation of selection biases by comparing the trigger decisions applied to the tag electron (Homogeneity Tests).

6.1 Quadrant Analysis

The Quadrant Analysis was developed to compare the decisions of two classifiers. Given a signal candidate and two classifiers, there are some possibilities: (i) both accept the candidate, (ii) both reject the candidate, or (iii) one classifier accepts, and the other rejects it.

The possible trigger decisions, referred to as quadrants, are used here to categorize events. The shower-shape variables employed in the offline electron likelihood selection are then compared across these quadrants to identify any potential bias introduced by the trigger step. Furthermore, in this analysis, it is also possible to evaluate the disagreement between two classifiers. The disagreement between two classifiers **A** and **B** can be defined as:

$$\text{disagreement} = \frac{N_{\mathbf{A}} + N_{\mathbf{B}}}{N_{\mathbf{A}} + N_{\mathbf{B}} + N_{\mathbf{AB}} + N_{\neg\mathbf{AB}}} \quad (4)$$

where $N_{\mathbf{A}}$ ($N_{\mathbf{B}}$) is the total number of candidates accepted exclusively by classifier **A** (**B**) and $N_{\mathbf{AB}}$ ($N_{\neg\mathbf{AB}}$) is the total number of candidates accepted (rejected) by both classifiers.

In this approach, only probe electron candidates selected using the $Z \rightarrow e^+e^-$ tag-and-probe method in collision data are considered. The offline selection applied to these candidates corresponds to the tight working point used as the trigger classifier. The results in Section 6.1.1 show the trigger selection performance using the tight requirement for duplicated triggers (Neural-Ringer and cut-based) during 2017.

6.1.1 Results

The shower-shape variables (R_η , R_ϕ , E_{ratio} , and R_{had}) [3] used in the offline likelihood selection are compared across the four trigger decision categories (quadrants). As shown in Figure 11, the disagreement—defined as cases where the classifiers make different decisions—is typically at the level of 1%. Here, only the results for $0.6 < |\eta| < 0.8$ and $35 < E_T < 40$ GeV are shown, but this behaviour is also observed for other regions of the $E_T \times |\eta|$ phase space. One should note that the working points of the two triggers are not

exactly the same, although the Neural-Ringer aims to achieve the same signal efficiency as the cut-based strategy and was operating as desired. In other words, the count differences between the blue and red profiles in Figure 11 seem to appear mainly due to small efficiency differences between the two triggers.

The shower-shape variables exhibit similar distributions across the four quadrants, although slight shifts between the blue and red profiles are visible in Figure 11. This is the case for R_η and R_ϕ , where the Neural-Ringer trigger consistently selects slightly more events in the region closer to one, which represents particles with slightly narrower showers in η and ϕ . Compared to the previously mentioned cases, the Neural-Ringer trigger's behaviour shows an even smaller effect in R_{had} , where tighter tails are observed. The E_{ratio} variable also shows that the Neural-Ringer trigger has a slight bias towards selecting more events in the tail region. A similar behaviour is seen in other regions of the $E_T \times |\eta|$ phase space.

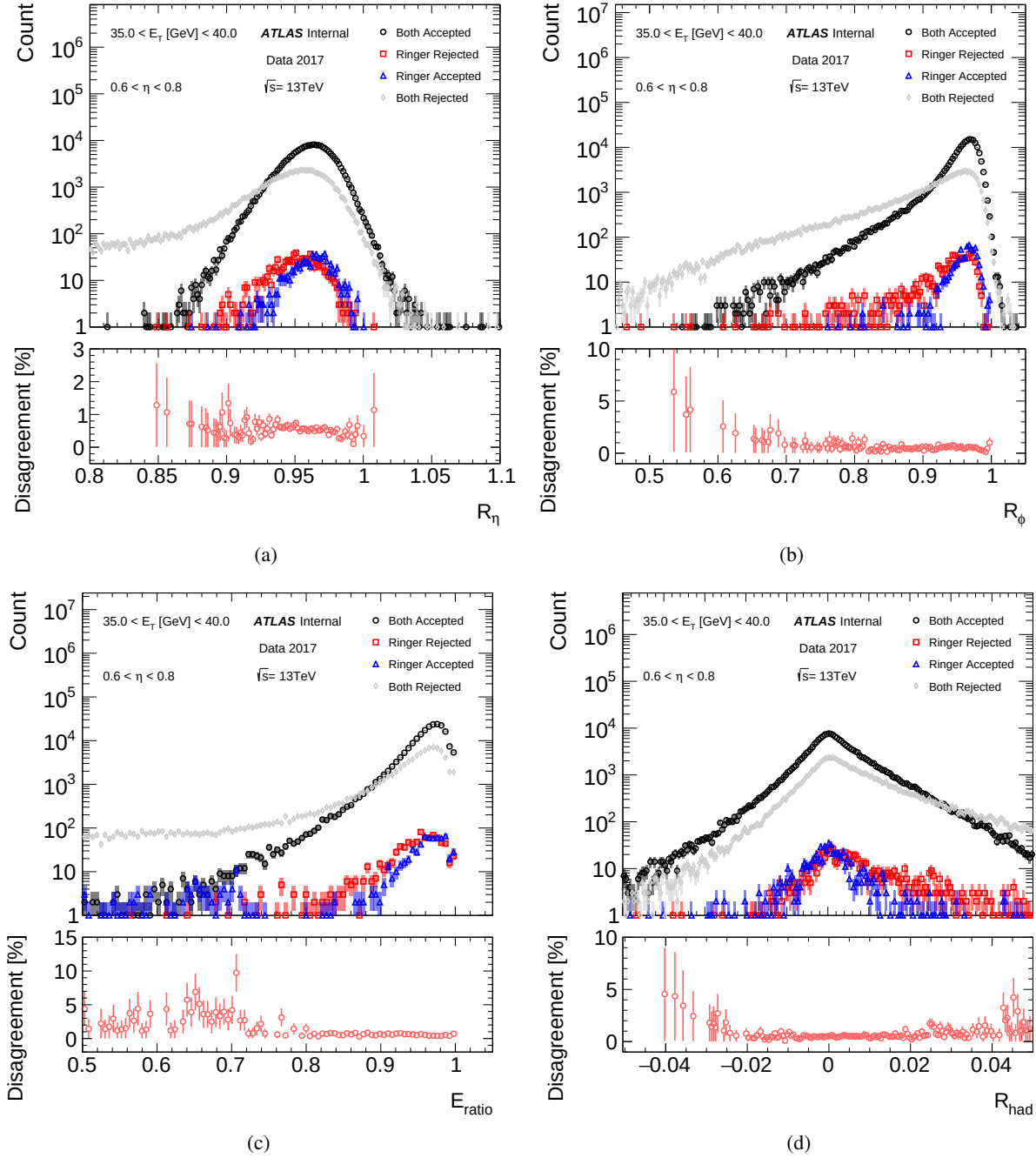


Figure 11: Quadrant Analysis plots for a selection of shower-shape variables used in the offline electron identification algorithm for the $0.6 < |\eta| < 0.8$ and $30 < E_T < 35$ GeV region. The top pad in each figure shows the raw number of candidates for the four mutually exclusive cases: accepted both with and without the Neural-Ringer (black); accepted only with the Neural-Ringer (blue); accepted only with the Cut-based (red); rejected both with Neural-Ringer and with Cut-based (grey). The bottom pad shows the disagreement as defined in Eq. (4). The absence of points of disagreement is due to the absence of events in the bins for the blue and red histogram cases.

6.2 Homogeneity Tests

The homogeneity tests evaluate the impact of using different trigger strategies—Neural-Ringer-based and cut-based—on the offline likelihood identification of tag electrons. To perform the study, a pair of duplicated single-electron triggers was used online. Both required $E_T > 28$ GeV, the ‘tight’ identification, and ‘loose’ isolation criteria. One version incorporated the Neural-Ringer-based selection, while the other did not. The same offline identification criteria used in the likelihood optimization described in [14] were applied throughout the analysis.

This analysis addresses this by applying homogeneity tests to histograms [23] to identify potential differences in the distributions of physics quantities arising from different trigger configurations. These tests are based on a test originally developed by Pearson [24] and popularly employed in many fields beyond high-energy physics, e.g. social sciences [25] and health [26], where they are usually called contingency or consistency tests.

To enable homogeneity tests without adapting them to ATLAS-specific setups, the 2017 collision data set was split into two statistically independent samples. These groups were constructed by alternating events taken at short time intervals to ensure similar data-taking conditions.

The p -value represents the probability of observing a result at least as extreme as the one measured, assuming the null hypothesis is true. In this case, the null hypothesis states that the distributions of the given variable are identical across the two samples. Comparing p -values across different samples helps assess whether the trigger configuration affects the shapes of the physics distributions.

To further investigate possible distortions introduced by the trigger algorithms, a bin-by-bin comparison is performed on histograms of shower-shape variables used in the offline electron identification. The signed χ -value, defined in Equation 5, is used for this purpose. In this expression, $r_{i,j}$ and $b_{i,j}$ are the event counts in the j^{th} bin and i^{th} $E_T \times |\eta|$ region, collected by the group using the trigger with and without the Neural-Ringer, respectively. This equation quantifies the deviation between the two trigger configurations and allows both positive and negative contributions to be visualized.

$$\chi_{i,j}^s = \frac{(r_{i,j} - b_{i,j})}{\sqrt{b_{i,j}}}, \quad (5)$$

6.2.1 Results

Regardless the trigger configuration, the homogeneity tests between the two groups result in similar p -values. The comparison of shower-shape distributions between the two statistically independent groups reveals fluctuations consistent with statistical uncertainty. No significant deviations are observed that would indicate systematic differences arising from the trigger configuration.

Figure 12 shows the residuals (bottom) for the R_η , R_ϕ , E_{ratio} , and R_{had} calorimetry variables when the reference histogram is obtained from collision data collected by the trigger without the Neural-Ringer for the first group (top). The residuals are dominated by statistical fluctuations (i.e., the residual values fluctuate around a central value) and are within expectations under the homogeneity hypothesis for most variables and regions. Similar behavior is observed for track variables.

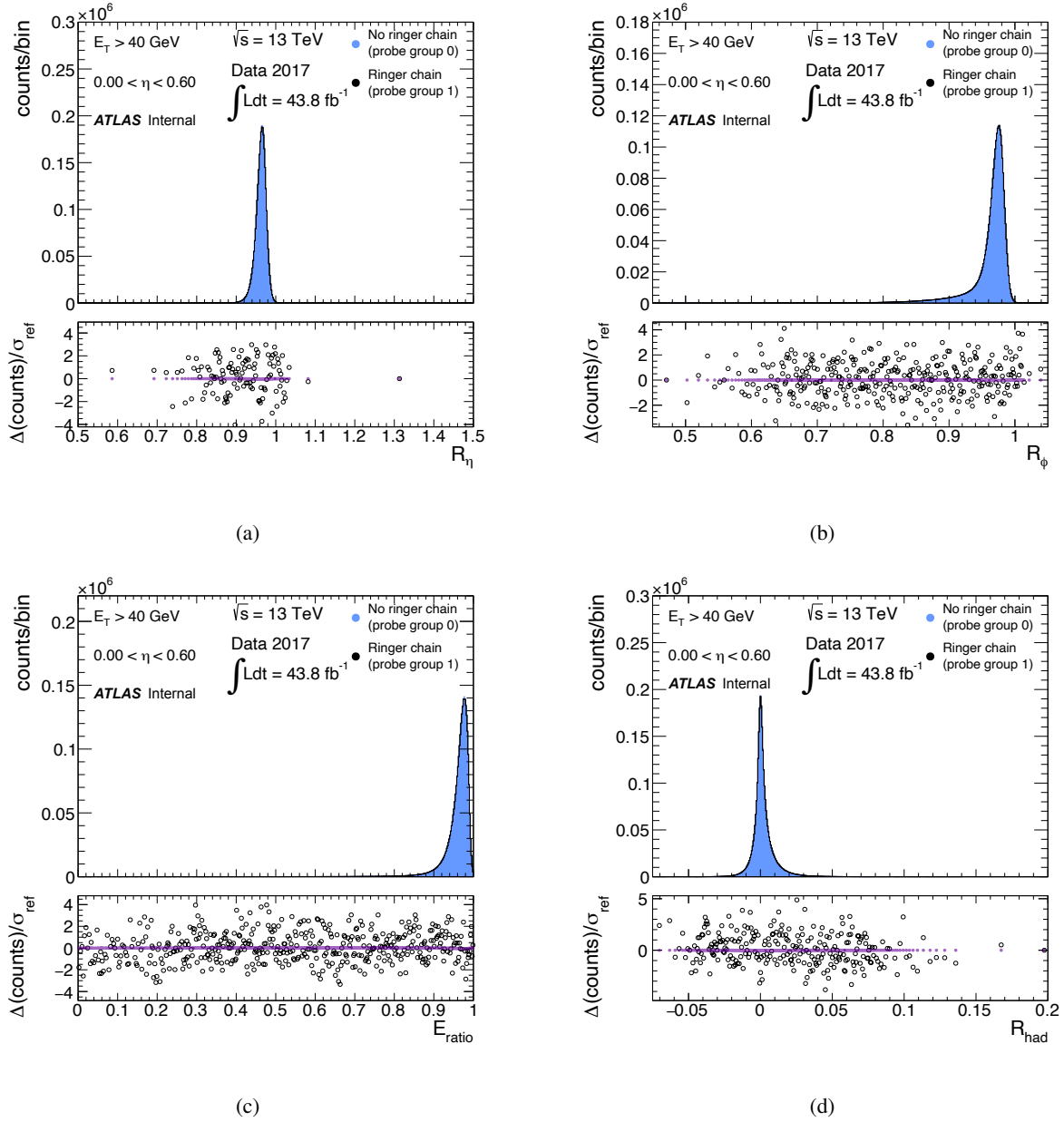


Figure 12: **Top:** Profile histograms of the R_η , R_ϕ , E_{ratio} , and R_{had} calorimetry variables used in the offline likelihood selection, shown for the $E_T > 40$ GeV and $0.0 < |\eta| < 0.8$ region. The distributions are obtained from two statistically independent data samples: one collected using the cut-based selection (blue area histogram), and the other using the trigger with the Neural-Ringer (black line histogram). The assignment of trigger type to group is arbitrary and alternated across time intervals to ensure statistical independence. **Bottom:** Bin-by-bin residuals computed using the signed χ^s statistic (black circles, see Eq. 5), which quantify the deviation between the two trigger configurations. The horizontal line at zero (purple dots) serves as a visual reference indicating no deviation.

7 Conclusion

An annular shower shape representation coupled to multi-layer perceptron neural networks (the Neural-Ringer) provides an innovative machine-learning-based design for online electron selection based on calorimetry information in the ATLAS experiment, aiming to meet specific trigger system requirements, such as reducing fake electrons.

For this, a feature extraction algorithm was implemented to produce concentric ring energy sums centered in the hottest cell of each calorimeter layer, which have been shown to provide an efficient description of calorimeter shower profiles and are fast computed through simple operations.

Neural-network models are employed to exploit non-linear correlations between rings, and the optimization of this strategy culminated in the deployment of a set of models specific to each region of pseudorapidity and transverse energy.

In the second half of 2017, the Neural-Ringer algorithm replaced a cut-based strategy in the FastCalo decision step of the HLT and became the baseline algorithm for triggering events containing electrons with E_T above 15 GeV, as part of the ATLAS online system's CPU-time reduction campaign. Furthermore, it was the first time a neural-network method was employed as a baseline event-selection algorithm in the ATLAS trigger system.

During the 2017 operation, the Neural-Ringer algorithm, trained on simulated data, reduced fake-electron trigger rates by a factor of 13.75 in the FastCalo and by a factor of 2 for the HLT final decision. At the same time, the Neural-Ringer achieved high electron-selection efficiencies very similar to those of relevant single-electron triggers without the Neural-Ringer. For 2018, the Neural-Ringer was trained with collision data, resulting in an estimated additional fake-rate reduction factor of 1.2–1.7 for fake-electron triggers in the FastCalo step with respect to 2017.

The Neural-Ringer selection was also studied using offline likelihood identification as a reference to obtain additional insights into its behavior by using two equivalent single-electron triggers with and without the Neural-Ringer. In this analysis, a high level of agreement was observed between the two triggers. Finally, shower-shape and tracking variables used in the offline likelihood identification were compared for electron data collected with the Neural-Ringer and cut-based triggers, and a strong statistical similarity was observed, indicating no bias in these variables introduced by the Neural-Ringer trigger configuration. This confirms that the Neural-Ringer trigger does not distort the inputs of the offline selection. As a result, the improved efficiency–rejection performance observed with the Neural-Ringer trigger can be attributed to its online behavior, without compromising the integrity of the offline identification.

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